



CS375 Spring 2026

Ralph Rostock

Week 1

Course Overview and Introduction to Software Engineering



Course Overview

- Course Number and Title: CS-375 – Software Engineering
- Credit Hours: 4
- Academic Term: Spring 2026
- Course Prerequisite: CS-371 Data Structures and Algorithms
- Class Meeting Location and Times: Thursday from 7:00 PM to 9:40 PM, PFA 107
- Office Hours
- Final Exam



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ACM on Software Engineering

“...software engineering requires both the analytical and descriptive tools developed in computer science and the rigor that the engineering disciplines bring to the reliability and trustworthiness of the systems that software developers design and implement while working cohesively in a team environment.”





ACM on Software Engineering (continued)

Software Engineering:

- Must be viewed as a discipline with stronger ties to computer science than it has to other engineering fields
- Must share common characteristics with other engineering disciplines, including quantitative measurement, structured decision making, effective use of tools, and reusable software components
- Must apply engineering methods and practices to the development of software, with special emphasis on the development of large software systems
- Must integrate the principles of mathematics and computer science with engineering methodologies to create extensible software
- Must include secure code standards and quality control concepts of manufacturing process design
- Must emphasize communication skills, teamwork skills, and professional principles and best practices





IEEE definition of Software Engineering

“Software engineering may be defined as the systematic design and development of software products and the management of the software process”





Software Engineering is Not Programming

- Writing Code vs Building Systems
- Scale, teams, longevity and risk



Why Software Systems Fail

Software Fails because Humans Fail

- Unclear or changing requirements (building the wrong system)
- Poor system design or architectural decisions
- Inadequate testing and quality assurance
- Ignoring non-functional requirements (scalability, reliability, security)
- Human and process issues: communication, coordination, and handoffs
- Accumulated technical debt and lack of maintenance



What is Software Engineering?

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